

LLM Multi-agents for Efficient Security in 6G Smart Factories

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Abstract—6G smart factories are hyperconnected cognitive manufacturing ecosystems that integrate massive robotics, IoT devices, and human-centric applications. Although promising, efficient system security, as highlighted in the recent 3GPP 6G workshop and 3GPP TR 22.870, faces two critical challenges: 1) *Fast adaptations*: Existing methods can hardly be adapted in time to the dynamic business logics generated by an AI-driven manufacturing system. 2) *Efficiency*: Large defense models consume excessive edge energy, while lightweight ones may be less effective. This motivates us to propose MAS-SEE, a novel LLM-based Multi-Agent System framework that balances Security and Energy Efficiency of 6G smart factories in practice. Our MAS-SEE consists of two key ingredients for multi-agent decisions and lightweight edge detection for dynamic security-energy optimization, respectively. Specifically, we introduce RuleGuard at the edge, a novel cognitive engine that generates adaptive policies by exploring a reasoning chain of *induction-deduction-analogy*. Such a chain-of-thought mechanism can properly tackle the first challenge for a highly dynamic AI-driven manufacturing system. Then, we present a novel confidence-based dynamic workload strategy, which directs computations to lightweight models or large ones, minimizing security energy consumption while maximizing the defense. Thus, it can properly address the second challenge. Experiments on real-world benchmarks across 10,000 samples show the effectiveness of our proposed MAS-SEE, yielding much higher defense, lower energy, and lower latency. Our studies potentially pave the way for the practical deployment of 6G smart manufacturing with efficient system security.

Index Terms—Multi-agent system, Large-language model, Cognitive rule engine, 6G Smart Factories, Efficient Security.

I. Introduction

A. Background

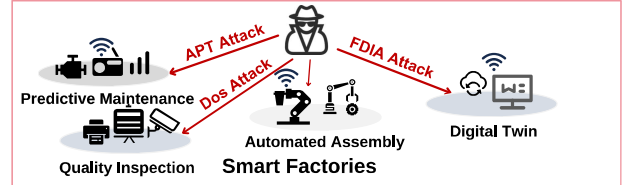
SMART factories—built on large-scale IoT deployments, AI-driven sensing, and cloud-edge-terminal collaboration—are recognised as a flagship 6G scenario for industrial automation [1]–[3]. Sixth-generation networks aim for Tbps-level throughput and sub-millisecond latency, enabling real-time control and digital-twin production lines [4]–[6]. Reflecting these ambitions, the 3GPP 6G Workshop [7] and Technical Report 22.870 [8] identify *security* and *energy efficiency* as parallel design goals for future systems.

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Strong LLM-based defense introduces high energy consumption



How to balance safety and energy?

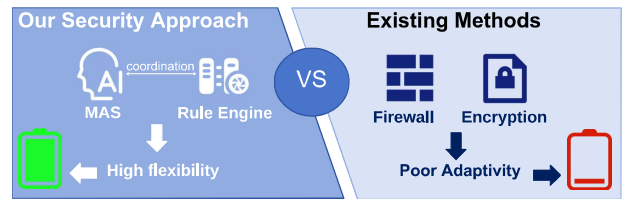


Fig. 1: High energy costs of existing defense approaches for securing 6G smart factories under highly dynamic environments.

The release of 3GPP TS 33.501 [9] relies on user-plane encryption and integrity to protect industrial endpoints. Although essential, these functions add extra processing and measurable power overhead on resource-constrained sensors. Edge-side solutions—such as compact neural networks combined with rule engines—reduce latency and backhaul traffic [19]–[21], [23]. However, the lack of standard robustness tests (e.g., ETSI SAI TST 007) and of cross-node rule synchronisation (e.g., OMG DMN) prevents edge nodes from adjusting policies as plant conditions change [10], [11]. Consequently, cloud-centric analytics and isolated edge rules remain siloed, and a single layer can hardly deliver both strong security and low power consumption. A cross-layer, intent-aware framework that converts high-level commands into coordinated security actions *and* energy-saving measures across devices, edge nodes, and the cloud is therefore required. A recent study [56] has also shown that optimizing semantic entropy can simultaneously improve transmission efficiency and channel security in wireless networks, indicating the feasibility of joint optimization strategies in next-generation industrial environments.

B. Motivation

Despite the mandates of 3GPP TS 33.501 [9] for pervasive encryption and integrity protection in industrial IoT, and the emergence of lightweight edge detection [19], [23], current practices in 6G smart factories remain siloed—static security policies on one side and high-energy cloud models on the other. This duality gives rise to three critical challenges:

1) **Heterogeneity–Security Gap**: Intent-driven [57], AI-

native management frameworks dynamically reconfigure devices and workflows, creating highly heterogeneous factory states that static rule engines cannot protect in real time.

2) **Energy–Detection Trade-off:** Large-model inference (e.g., Llama2-7B) offers strong detection but consumes hundreds of joules per query, while compact models (e.g., CMSIS-NN) cut energy by over 80% yet drop accuracy to around 68% [26], [27].

3) **Lack of Real-Time Policy Orchestration:** Existing intent-based frameworks incur parsing delays above 500 ms and cannot coordinate decisions across edge detectors, rule engines, and cloud controllers with sub-10 ms responsiveness [35], [37].

These limitations motivate a cross-layer, multi-agent solution that fuses intent parsing, lightweight edge perception, and global coordination [38]–[40]. To this end, we propose MAS-SEE, a three-layer multi-agent system framework for dynamic energy–security co-optimization in 6G smart factories.

C. Our Method

The MAS-SEE framework orchestrates a seamless interplay between perception, detection, and policy synthesis across cloud, edge, and terminal layers. It balances energy consumption and security performance by dynamically adapting to real-time telemetry and operator intents. Built on principles of modularity and multi-agent collaboration, MAS-SEE supports flexible deployment, continuous refinement, and closed-loop optimization. We distill its capabilities into five key attributes:

- 1) **Lightweight:** IoT devices continuously gather real-time metrics, perform distributed preprocessing, and minimize transmission overhead.
- 2) **Adaptive:** Each edge node ingests local telemetry, computes normalized anomaly scores, and adjusts detection thresholds on the fly [44].
- 3) **Intent-driven:** A cloud-hosted parser converts natural-language instructions into cross-layer defense policies via multi-agent reinforcement learning [45].
- 4) **Collaborative:** All agents share a centralized policy pool, enabling knowledge transfer between normal monitoring and intent-driven modes.
- 5) **Efficient:** The framework dynamically switches between compact on-device models and cloud agents, conserving energy while maintaining high detection accuracy.

Keeping these goals in mind, we first present a MAS-SEE prototype in a 6G smart-factory testbed, integrating distributed IoT agents, hybrid edge detectors, and a cloud-based intent parser. The prototype also includes an energy-aware scheduler that dynamically shifts inference workloads between edge and cloud, and a closed-loop feedback mechanism for policy refinement. Finally, MAS-SEE outputs fine-grained defense and power-saving directives to factory controllers for real-time enforcement. Experiments on real industrial IoT traces and a hardware-in-the-loop pilot demonstrate the effectiveness of MAS-SEE: achieving high detection accuracy while reducing edge energy consumption. *We will release our implementation to contribute to the community.*

D. Main Contributions

In summary, the contributions of this paper are fourfold:

- **MAS-SEE Framework:** We present *MAS-SEE*, the first intent-driven, multi-agent architecture for joint security–energy optimisation for 6G smart factories. Our MAS-SEE integrates lightweight edge detection, cognitive rule reasoning, and cloud-level policy coordination in an end-to-end fashion.
- **RuleGuard Engine:** We develop *RuleGuard*, a novel cognitive rule engine that rely on a three-stage induction–deduction–analogy pipeline to generate strategies. An ADMM-based optimiser continuously updates rule weights, enabling fast adaptation to evolving industrial threats.
- **LLM Multi-Agents:** We introduce a series of LLM-powered intent agents for energy, security, and decision, respectively. This quartet achieves sub-10ms intent parsing and real-time cross-layer policy orchestration, thus bridging the semantic gap between user directives and low-level defence actions.
- **Extensive Experiments:** We build a hardware-in-the-loop 6G smart-factory testbed, and introduce a comprehensive dataset, covering 10,000 samples with mixed benign and attack traffic. Extensive experiments demonstrate 91.7% decision accuracy and a 23.7% reduction in edge energy consumption, confirming the effectiveness of the proposed MAS-SEE.

II. Related Work

A. Cloud–Edge–End Architectures:

Hierarchical and fog–edge–cloud frameworks balance latency, bandwidth, and computational resources in industrial networks [46], [47]. Hybrid schemes further coordinate edge and cloud nodes to reduce delay and improve resource utilisation [20], [48]. *Key difference:* To the best of our knowledge, the proposed MAS-SEE is the first intent-driven framework for joint security-and-energy optimisation in 6G smart-factory scenarios.

B. Lightweight Models and Rule Engines:

TinyML and compact DNNs enable on-device anomaly detection under tight power budgets [12], [27], [29], yet their robustness against sophisticated industrial attacks is limited. Static rule engines such as SNORT and Esper CEP process known threats efficiently but lack runtime adaptability and distributed scalability [31], [32], [34]. Key differences are:

- We introduce RuleGuard, a novel cognitive rule engine that adaptively fuses multimodal reasoning with ADMM-based adaptive rule updating.
- The proposed MAS-SEE couples RuleGuard with lightweight edge models in a single pipeline, boosting detection accuracy while maintaining low energy consumption.

C. Intent-Driven and Multi-Agent Systems:

Intent-Based Networking (IBN) and Multi-Agent Reinforcement Learning (MARL) automate high-level policy enforcement and distributed decision-making [35], [37], [49],

[53]. Existing solutions, however, incur parsing latencies exceeding 500 ms and struggle to bridge semantic gaps between user intents and device-level actions, making sub-10 ms cross-layer coordination impractical. Two key differences include

- The proposed MAS-SEE achieves sub-10 ms intent parsing via an LLM-based Intent Agent.
- Four specialised agents—Intent, Energy, Security, and Decision—operate in a unified, closed loop to deliver real-time security-and-energy co-optimisation across terminal, edge, and cloud layers.

III. Overall Framework

In 6G-enabled smart meter scenarios, traditional centralized security protection and energy management models face challenges such as insufficient real-time responsiveness, rigid strategies, and resource constraints at the edge. To address the collaborative optimization of dynamically evolving security threats and terminal energy efficiency, we present MAS-SEE, a “Terminal-Edge-Cloud” three-tier collaborative architecture (Fig. 2):

- **Terminal Layer:** Equipped with embedded sensors, terminals perform real-time data acquisition including voltage, current, communication patterns, and device-level abnormalities. This data serves as raw input for edge intelligence.
- **Edge Layer:** Lightweight AI detection models and RuleGuard are deployed for localized real-time decision-making and low-latency responses.
- **Cloud Layer:** A cognitive multi-agent system performs global optimization. It includes the *Intent Agent (IA)*, the *Energy Agent (EA)*, the *Decision Agent (DA)*, and the *Security Agent (SA)*—each managing high-level parsing, multi-objective learning, and policy consistency, respectively.

This design forms a closed-loop cycle: Terminal Perception

→ Edge Inference → Cloud Optimization → Policy Feedback. The system achieves context-aware, user-intent-aligned, and resource-conscious behavior across layers.

A. Edge Layer: Lightweight AI and RuleGuard

The edge layer adopts a dual-core design combining a lightweight AI model and a RuleGuard reasoning module. Together, they enable real-time, resource-efficient decisions by converting statistical patterns into actionable insights:

- 1) **Lightweight AI Model:** Captures protocol-aware anomalies and encodes them into semantic labels (e.g., “low voltage + burst traffic → suspected DoS”) within sub-millisecond latency. These labels serve as inputs for rule reasoning.
- 2) **RuleGuard:** Mimics cognitive reasoning through *Induction*, *Deduction*, and *Analogy*. Eight scoring modules assess the input, and an ADMM-based optimizer fuses their outputs for adaptive decision-making.

This design ensures fast, interpretable, and energy-efficient decisions at the edge. The AI model handles scalable detection, while RuleGuard refines decisions for accuracy and context awareness.

B. Cloud Layer: Multi-Agent Collaboration and Closed Control

When either of the following occurs—(1) edge confidence drops below a predefined threshold, or (2) a user issues a natural language intent—the system triggers cloud-level reasoning via its multi-agent system:

- 1) **Intent Agent (IA):** Translates natural language commands (e.g., “Prioritize grid stability during peak hours”) into actionable policy weights. Combines semantic parsing with historical operation patterns to balance security-energy priorities.
- 2) **Energy Agent (EA):** Optimizes power consumption by dynamically adjusting device sleep cycles, communication

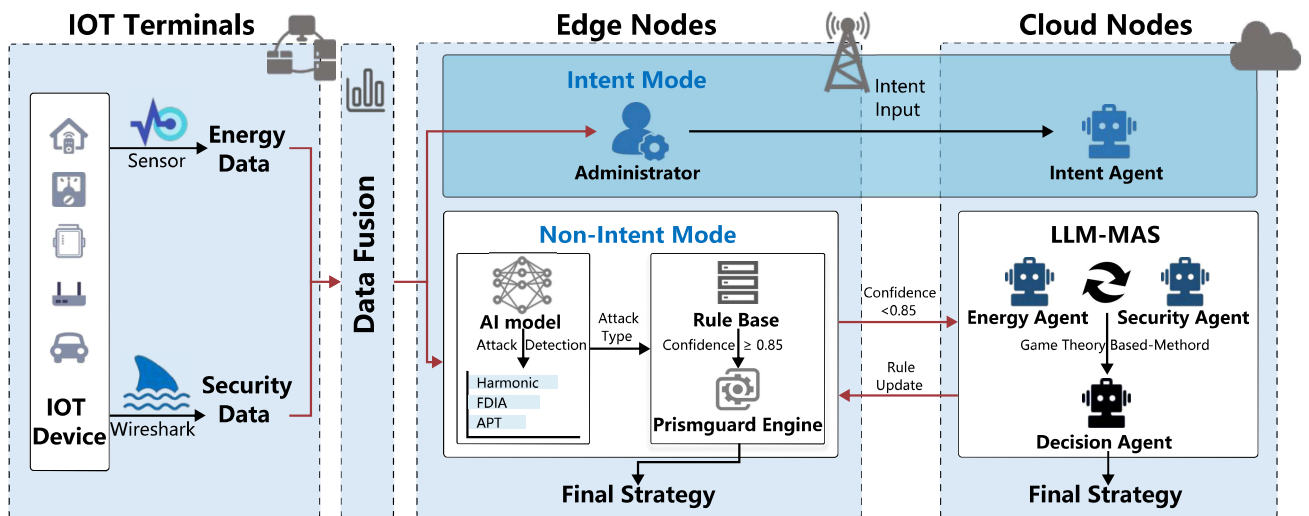


Fig. 2: MAS-SEE integrates the IoT Terminal Layer for real-time collection, Edge Node Layer with lightweight AI for rapid threat detection and RuleGuard-driven decisions, and Cloud MAS Layer for intent parsing and global policy optimization.

rates, and sensor sampling frequencies based on real-time grid load. Maintains adaptive baselines to minimize security overhead.

3) **Security Agent (SA)**: Evaluates attack risks through cross-layer threat modeling, activating targeted defenses like traffic filtering or hardware isolation. Implements risk-adaptive control with failsafe mechanisms for critical infrastructure.

4) **Decision Agent (DA)**: Orchestrates global optimization through a leader-follower game: DA sets security constraints while EA/SA compete for resource allocation, with IA ensuring human intent alignment.

The cloud MAS updates edge strategies and rule bases continuously, forming a closed-loop cognitive architecture. MAS-SEE thus functions as a distributed organism—learning, evolving, and defending in accordance with user intent and system context.

IV. The Proposed RuleGuard

This section focuses on a specific component, namely the cognitive-driven multi-modal adaptive security and energy-efficiency rule engine (Fig. 3). The content here centers around elaborating on this particular engine in detail.

A. Induction: Dynamic Candidate Rule Generation

Perform rapid co-occurrence statistics on historical multimodal labels (power deviations, IP interactions, time contexts) to select high-frequency, low-redundancy combinations as candidate rule antecedents.

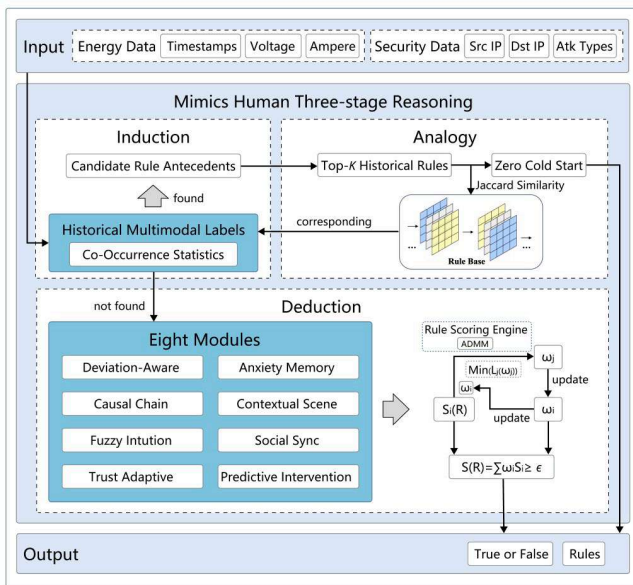


Fig. 3: Edge-deployed RuleGuard architecture: processes multimodal data through three cognitive phases—Induction, Deduction, and Analogy—to adaptively generate rules.

RuleGuard mimics human three-stage reasoning—**Induction**, **Deduction**, and **Analogy**—to process multimodal data such as timestamps, voltage, current, and attack types. In the Induction phase, co-occurrence statistics

on historical multimodal labels generate candidate rule antecedents. The Deduction phase evaluates these candidates using eight parallel cognitive modules, each scoring anomalies based on different contextual or statistical criteria. Scores are fused by an ADMM-based rule scoring engine to determine final outputs. The Analogy phase enables zero cold start by transferring top-K historical rules with the highest Jaccard similarity, ensuring adaptive and context-aware edge rule generation.

To quantify the relevance of candidate rules, co-occurrence confidence and information gain are computed as:

$$\text{Conf}(A \Rightarrow B) = \frac{\text{Supp}(A \cap B)}{\text{Supp}(A)} \quad (1)$$

$$\text{IG}(A) = H(B) - H(B|A) \quad (2)$$

where A represents feature combinations (e.g., power spike + IP port scan), and B is the outcome label.

B. Deduction: Eight-Module Scoring

Upon arrival of new data, eight sub-modules run in parallel, each returning a normalized score $S_i \in [0, 1]$. The modules are defined as follows:

- **Deviation-Aware**: Detects real-time power deviation using historical mean μ_P and standard deviation σ_P .
- **Causal Chain**: Scores multi-step event chains (e.g., connect→download→power spike) with time intervals Δt_i and power change ΔP .
- **Contextual Scene**: Applies a scene-specific threshold θ_s based on the current business mode or time period.
- **Fuzzy Intuition**: Combines Shannon entropy $H(P_{t-\tau:t})$ of power signals with IP-pattern similarity sim_{IP} .
- **Anxiety Memory**: Accumulates minor anomalies via exponential-decay memory $\mathcal{A}_d(t)$, increasing sensitivity to consecutive events.
- **Social Sync**: Detects synchronised fluctuations across multiple devices, revealing distributed attack patterns.
- **Trust Adaptive**: Updates trust level $T(t)$ from past positive/negative behavior, and adjusts thresholds on the fly.
- **Predictive Intervention**: Uses an ARIMA model to forecast future power $\hat{P}(t + \Delta)$ and assign proactive scores.

The fused decision score is:

$$S(R) = \sum_{i=1}^8 \omega_i S_i(R), \quad \sum_{i=1}^8 \omega_i = 1. \quad (3)$$

Core scoring formulas are summarised in Table I, and all symbol definitions are listed in Table II.

C. Analogy: Rule Transfer and Maintenance

When new devices or attack types emerge, compute Jaccard similarity between label sets to transfer the Top-K highest-similarity historical rules, enabling zero cold start.

To identify transferable rules, we compute the Jaccard similarity between current and historical label sets:

$$\text{Sim}_{\text{Jaccard}}(R_a, R_b) = \frac{|R_a \cap R_b|}{|R_a \cup R_b|} \quad (4)$$

TABLE I: Core scoring formulas of the eight submodules

Module	Scoring Formula
Deviation-Aware	$S_{\text{dev}} = \exp(-\frac{ P(t)-\mu_P }{k\sigma_P}) \mathbf{1}_{J < \delta}$
Causal Chain	$S_{\text{causal}} = \exp(-\alpha_1 \Delta t_1 - \alpha_2 \Delta t_2 - \alpha_3 \Delta P)$
Contextual Scene	$S_{\text{ctx}} = \mathbf{1}_{C(t)=s} \exp(-\frac{ P(t)-\theta_s }{\theta_s})$
Fuzzy Intuition	$S_{\text{fuz}} = H(P_{t-\tau:t}) (1 - \text{sim}_{\text{IP}})$
Anxiety Memory	$\mathcal{A}_d(t) = \gamma \mathcal{A}_d(t-1) + (1-\gamma)I_{\text{ex}}, S_{\text{anx}} = \frac{\mathcal{A}_d(t)}{\tau}$
Social Sync	$S_{\text{soc}} = \frac{1}{n} \sum_{d=1}^n \mathbf{1}_{\Delta I_d > \delta}$
Trust Adaptive	$\begin{aligned} T(t+1) &= T(t) + \eta^+ p - \eta^- n, \\ S_{\text{trust}} &= \frac{P(t)-\mu_P}{\sigma_P} (1 - T(t)) \end{aligned}$
Predictive Intervention	$\begin{aligned} \hat{P}(t+\Delta) &= \text{ARIMA}(P_{t-n:t}), \\ S_{\text{pred}} &= \exp(-\beta \max(0, \hat{P} - P_{\text{crit}})) \end{aligned}$

TABLE II: List of Parameters

Symbol	Description
S	number of rules to transfer
R	available system resources
τ	sliding window size
ϵ	decision threshold
i	index of the i th submodule
ω	fusion weight vector
ρ	ADMM penalty parameter
$\mathcal{L}_j$	local loss of module j
$H(\cdot)$	Shannon entropy
$\delta, J, k, \alpha_i, \beta, \gamma, \eta^+, \eta^-$	tuning parameters

Top- K rules with the highest similarity are transferred to support rule generation in previously unseen contexts.

D. Weight Fusion via Distributed ADMM

Define each module's local loss $\mathcal{L}_j(\omega_j)$ measuring false alarms and misses, and solve

$$\min_{\{\omega_j\}, z} \sum_{j=1}^8 \mathcal{L}_j(\omega_j) \quad \text{s.t.} \quad \omega_j = z, \forall j \quad (5)$$

using ADMM to ensure global consistency and local optimality with online updates. Each module optimizes its local weight vector ω_j by minimizing a penalized objective:

$$\omega_j^{(t+1)} := \arg \min_{\omega_j} \left(\mathcal{L}_j(\omega_j) + \frac{\rho}{2} \left\| \omega_j - z^{(t)} + u_j^{(t)} \right\|^2 \right) \quad (6)$$

The global consensus variable z is updated by averaging local estimates and dual variables:

$$z^{(t+1)} := \frac{1}{N} \sum_{j=1}^N \left(\omega_j^{(t+1)} + u_j^{(t)} \right) \quad (7)$$

Dual variables u_j are updated accordingly:

$$u_j^{(t+1)} := u_j^{(t)} + \omega_j^{(t+1)} - z^{(t+1)} \quad (8)$$

E. End-to-End Workflow

We integrate the above three phases into an online workflow that executes continuously on edge nodes. The complete procedure is outlined in Algorithm 1.

Algorithm 1 Cognitive-Driven Rule Engine Workflow

Require: historical context $\{c_t\}$, window τ , threshold ϵ , transfer count K
Ensure: system actions
1: **Induction:** generate candidate rule set R
2: **Initialize:** compute $\mathcal{L}_j$ and solve ADMM for ω
3: **while** system is online **do**
4: Data preprocessing: read and smooth (V, I, P) , update (μ_P, σ_P) , construct context
5: Multi-module scoring: compute S_i and fuse; if $S(R) \geq \epsilon$, execute action
6: Logging and update: record logs, periodically rerun ADMM to update ω
7: Analogy maintenance: transfer Top- K rules for new scenario, prune outdated, induce new candidates
8: **end while**

V. The Proposed Cloud-Layer Multi-Agent System

To overcome the limitations of conventional IBN in dynamic 6G environments, we propose a cloud-based Multi-Agent System (MAS) that balances energy efficiency and security at the edge (Fig. 4). At its core, a biologically inspired Intent Agent manages the entire intent lifecycle through a four-tier cognitive model: Neural Loop for semantic parsing, Immune Loop for conflict detection, Social Coordinator for utility-driven optimization, and MetaCell for abstraction evolution. Coordinated via a dialogue framework, the MAS includes four agents: Intent, Security, Energy, and Decision, which collaboratively transform high-level directives into adaptive, utility-aware policies.

A. Intent Agent: Cognitive Architecture

1) *Transformer-Based Semantic Parsing:* The Neural Loop functions analogously to the cerebral cortex in biological systems, serving as the central interpreter for natural language intents. Its Transformer-based encoder classifies each intent I into a predefined semantic category set S :

$$\hat{s} = \arg \max_{s \in S} P(s|I; \theta), \quad (9)$$

where θ denotes the learnable parameters of the model. In addition to classification, the encoder generates high-dimensional embeddings $e_I = f_{\text{enc}}(I; \theta)$, enabling semantic similarity computation and downstream policy generation, effectively acting as a semantic reflex arc for intent processing.

2) *Anomaly-Aware Policy Verification:* To ensure robustness against policy conflicts, the Immune Loop emulates adaptive immunity by monitoring policy vectors $\mathbf{p}$ and identifying anomalies. Policy representations are modeled as samples from a multivariate Gaussian distribution $\mathcal{N}(\mu, \Sigma)$, and the Mahalanobis distance is employed for conflict detection:

$$\text{ConflictScore} = (\mathbf{p} - \mu)^\top \Sigma^{-1} (\mathbf{p} - \mu). \quad (10)$$

where μ means vector of policy distribution is derived from historical data. When the conflict score exceeds a dynamic threshold τ , the system initiates automated countermeasures

such as rollback, quarantine, or regeneration to maintain policy integrity.

3) *Distributed Resource Optimization*: Inspired by self-organizing behavior in biological societies, the Social Coordinator formulates resource scheduling as a utility-maximizing optimization problem:

$$\max_{\mathbf{x}} \sum_{i=1}^n u_i(x_i), \quad \text{s.t.} \quad \sum_{i=1}^n x_i \leq R, \quad x_i \geq 0, \quad (11)$$

where x_i is the allocated resource to user i , u_i the utility function, and R the total capacity. A distributed primal-dual method is used to ensure fair and stable allocation across users, reflecting biological homeostasis. The optimization is solved via a distributed primal-dual algorithm. The Lagrangian function is:

$$\mathcal{L}(\mathbf{x}, \lambda) = \sum_{i=1}^n u_i(x_i) + \lambda \left(R - \sum_{i=1}^n x_i \right) \quad (12)$$

Each agent updates its local variable using:

$$x_i^{(t+1)} = \arg \max_{x_i \geq 0} [u_i(x_i) - \lambda^{(t)} x_i] \quad (13)$$

The dual variable λ is updated as:

$$\lambda^{(t+1)} = \left[\lambda^{(t)} - \eta \left(R - \sum_{i=1}^n x_i^{(t)} \right) \right]^+ \quad (14)$$

where $\eta > 0$ is the learning rate.

4) *Abstraction and Concept Drift Adaptation*: The Meta-Cell component serves as a long-term semantic memory inspired by stem cell differentiation. It dynamically clusters embeddings e_i into k semantic prototypes using incremental k -means:

$$C = \arg \min_C \sum_{j=1}^k \sum_{x \in S_j} \|x - c_j\|^2, \quad (15)$$

where c_j denotes the centroid of cluster S_j . This enables efficient intent generalization, few-shot adaptation, and continuous learning. Cluster drift detection is incorporated to capture evolving intent semantics over time.

B. MAS Architecture and Workflow

1) *Operational Triggers and Workflow Overview*: The MAS operates in standby mode and is activated only under specific conditions:

- **Low Confidence from Edge AI**: When the edge AI model produces a confidence score lower than 70%, indicating a complex or uncertain scenario.
- **Administrator Intent Input**: When human operators input directives that explicitly shift the system's operating priorities, such as prioritizing energy saving over security, or vice versa.

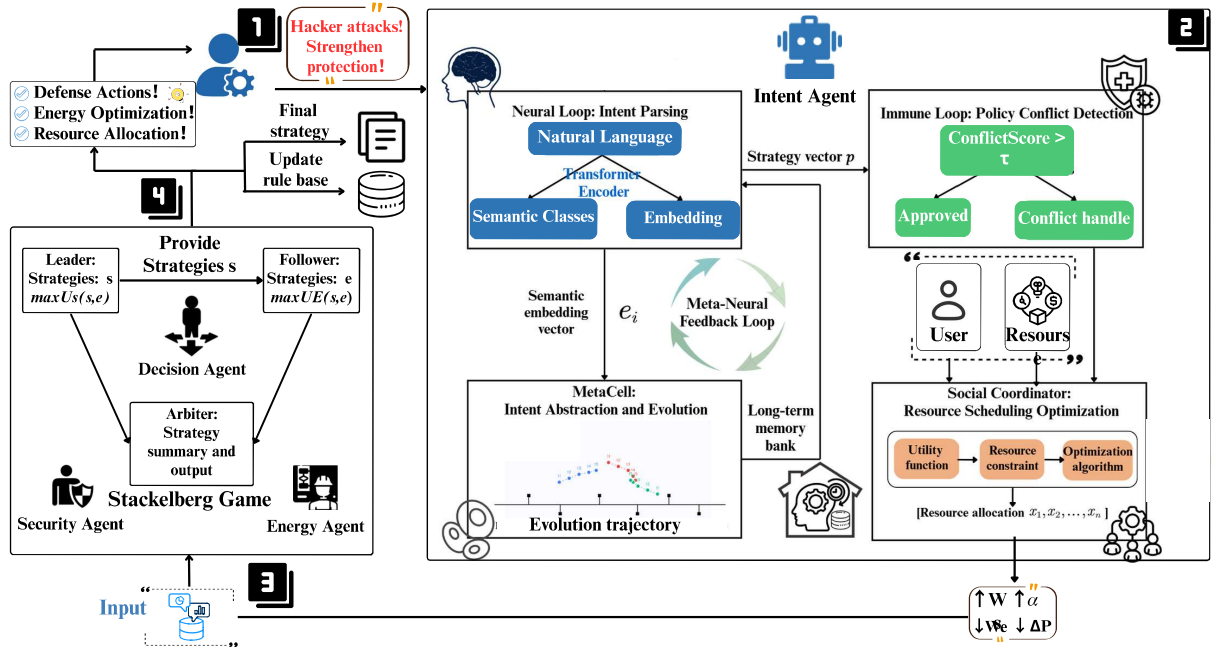


Fig. 4: Intent-driven MAS collaboration: The Intent Agent (IA) translates user directives into system priorities, while Security (SA), Energy (EA), and Decision Agents (DA) jointly optimize security-energy trade-offs via Stackelberg game-theoretic coordination within the terminal-edge-cloud architecture.

Algorithm 2 Intent Parsing and Semantic Verification

Require: Intent I , semantic categories $\mathcal{S}$, similarity threshold δ

Ensure: Category $\hat{s}$, verified parameters P

- 1: **Encode intent:**
 $\mathbf{h}_I \leftarrow \text{T5Encoder}(I)$
- 2: **Predict semantic category:**
 $\hat{s} \leftarrow \arg \max_{s \in \mathcal{S}} \text{softmax}(\mathbf{W}\mathbf{h}_I + b)$
- 3: **Generate candidate parameters:**
 $C \leftarrow \{p_1, p_2, \dots\}$
- 4: **for each** $p \in C$ **do**
- 5: **Encode parameter:**
 $e_P \leftarrow f_P(p)$
- 6: **Compute similarity:**
 $\text{sim}(I, p) = \frac{e_I \cdot e_P}{\|e_I\| \cdot \|e_P\|}$
- 7: **end for**
- 8: **Select most relevant parameter:**
 $p^* \leftarrow \arg \max_{p \in C} \text{sim}(I, p)$
- 9: **if** $\text{sim}(I, p^*) < \delta$ **then**
- 10: **Trigger human-in-the-loop verification**
- 11: **end if**
- 12: **Set final parameter:**
 $P \leftarrow p^*$
- 13: **return** $\hat{s}, P$

During normal operations, edge AI models continuously monitor power usage, voltage, current, and IP-level traffic to handle straightforward situations locally. If threats are detected with high confidence, decisions are made via rule-based logic at the edge. Otherwise, control is escalated to the MAS, which performs fine-grained decision-making in the cloud, particularly when handling trade-offs between conflicting objectives.

2) *Agent Roles and Collaboration Mechanism:* The MAS is composed of four specialized agents—each with a distinct responsibility but coordinated via a dialogue-based multi-agent framework.

- **Energy Agent:** Monitors energy consumption metrics and dynamically adjusts system-level power-saving strategies to reduce usage without compromising operation.
- **Security Agent:** Continuously analyzes network traffic and system behavior to detect abnormal activities. It proposes countermeasures based on threat levels, adapting its model to new vulnerabilities.
- **Decision Agent:** Acts as the central coordinator, aggregating the outputs from the Energy and Security agents, resolving trade-offs, and generating actionable system strategies. It also manages rule base updates.
- **Intent Agent:** Interprets administrator directives and modifies the system's internal preference weights, influencing the optimization objectives dynamically.

These agents interact through Microsoft's AutoGen multi-agent dialogue framework, enabling goal-oriented negotiation among agents with conflicting utilities. The framework ensures convergence to a balanced strategy under multi-objective constraints.

3) *Game-Theoretic Decision Model:* The collaboration between the Energy and Security agents is modeled using a Stackelberg game, reflecting the system's hierarchical optimization nature. The Security agent acts as the leader due to the higher priority given to safety, and the Energy agent acts as the follower. They interact under the constraint:

$$s + e \leq 1 \quad (16)$$

where s is the security effort, and e is the energy-saving effort. Both represent proportions of the total system resources.

The utility functions are defined as:

$$U_E(s, e) = -[P_0(1 - e) + \Delta P \cdot g(s)] \quad (17)$$

$$U_S(s, e) = -R_0 \exp[-\alpha s(1 - e)] \quad (18)$$

where U_E and U_S represent the utilities of the Energy and Security agents, respectively. Here, P_0 is the baseline power consumption, ΔP is the additional power cost due to security measures, $g(s)$ describes how security efforts impact power consumption, R_0 is the baseline security risk, and α is the risk mitigation coefficient indicating the effectiveness of security measures.

The Energy agent best-responds with $e^*(s) = 1 - s$, while the Security agent chooses s to maximize U_S accordingly. This Stackelberg setup reflects real-world operational priorities where over-optimizing for energy may weaken security.

4) *Intent-Driven Utility Adjustment:* When administrators input high-level goals (e.g., "prioritize energy saving this weekend"), the Intent Agent modifies the utility functions by introducing weighted preference terms:

$$U_E = -[P_0(1 - e) + \Delta P g(s)] + w_E H_E \quad (19)$$

$$U_S = -R_0 \exp[-\alpha s(1 - e)] + w_S H_S \quad (20)$$

Here, w_E and w_S reflect the administrator's emphasis on energy efficiency or security, respectively. H_E and H_S represent secondary objectives, such as maintenance cost or long-term reliability.

The new optimal energy-security trade-off is computed as:

$$s^* = \arg \max_{s \in [0,1]} U_S(s, 1 - s) + w_S H_S(s) \quad (21)$$

By adjusting w_E and w_S , administrators influence the system's preference toward energy or security, effectively reconfiguring the MAS equilibrium strategy. This adjustment shifts the Stackelberg equilibrium, enabling flexible responses to administrator preferences.

5) *Strategy Output and Rule Evolution:* The finalized strategy coordinates actions such as:

- **Security side:** Activating adaptive defenses like traffic filtering and hardware isolation while quarantining anomalous traffic patterns.

• **Energy side:** Implementing dynamic power reduction through transmission throttling and device sleep modes to maximize operational efficiency.

Validated strategies are abstracted into new rules and integrated into the edge rule engine. Each rule undergoes consistency checks to avoid conflicts and ensure measurable performance gains. Table III shows examples of learned rules.

TABLE III: Example of Rule Updates from MAS Decisions

Scenario	Strategy	Confidence	Added
Security Priority	Increase Defense Level	95%	Yes
Energy Priority	Reduce Power Consumption	92%	Yes

This feedback loop acts as a catalyst, enhancing the system's adaptive policy evolution, continuously refining strategies in response to operational scenarios.

VI. Experiments

A. Experimental Setup

The experimental platform consists of several x86 servers and a single NVIDIA Jetson Nano, each playing distinct roles in simulating a smart factory IoT system. Servers emulate terminal nodes (smart meters), an edge node, and a cloud node. The Jetson Nano serves as an active attack injector.

- **Terminal nodes:** Virtualized processes or containers on servers emulate smart meters, periodically generating voltage, current, and power data, and sending them to the edge node.
- **Edge base station node:** A server runs the attack detection model and rule engine, forwarding selected data to the cloud.
- **Cloud core network node:** Hosts the multi-agent reinforcement learning system and intent parser.
- **Attack injection node:** The Jetson Nano injects FDIA, DoS, harmonic, APT, and poisoning attacks, targeting both terminals and the edge, covering various protocols and scenarios.

Devices synchronize time using NTP to maintain consistent timestamps across the system.

B. Datasets

The dataset combines virtual terminal simulation and physical attack injection. Each terminal periodically generates synthetic voltage, current, and power data, while the Jetson

Nano injects abnormal packets to simulate attack scenarios. This hybrid setup yields realistic traces with scalable coverage. Each sample is a structured record with the following fields:

TABLE IV: Sample fields of the constructed dataset.

Field	Description
Voltage (V)	Simulated or injected voltage
Current (A)	Simulated or injected current
Power (W)	Simulated or injected power
Timestamp	UTC timestamp
Source IP	Terminal or attacker IP
Destination IP	Edge or cloud node IP
Attack Type	None / FDIA / DoS / Harmonic / APT / Poisoning

The dataset includes 10,000 samples spanning normal and four attack categories, supporting training and evaluation of decision modules and enables robust evaluation across normal and attack scenarios.

C. Baselines

Baseline comparisons are organized by system mode and component granularity, as follows:

1) Full System Comparison in Non-Intent Mode:

The system combines a Realguard DNN and custom rule engine at the edge. When rule confidence falls below 0.85, data is offloaded to a DDPG-based MAS in the cloud. The following baselines are considered:

- **Rule-only:** Drools v8.1 with 67 handcrafted rules.
- **AI-only:** MobileNetV2 deployed at the edge.
- **Cloud-only:** Raw data sent to Llama3-70B for end-to-end inference.

2) Component Replacement in Non-Intent Mode:

To assess modular robustness, we replace key components:

- **Rule Engine:** Snort IDS v3.0 and Esper CEP v9.0.
- **MAS:** DQN (no coordination), MetaGPT v0.8.2 (prompt-based), ChatGPT-4o (structured prompts, no RL).

3) Full System Comparison in Intent Mode:

In this mode, natural language instructions are parsed by Llama3-8B into goal vectors processed by the MAS. Baselines include:

- **GPT-4 Turbo:** Directly outputs JSON-formatted policies.
- **DeepSeek-67B:** Performs context-based decisions without agents.

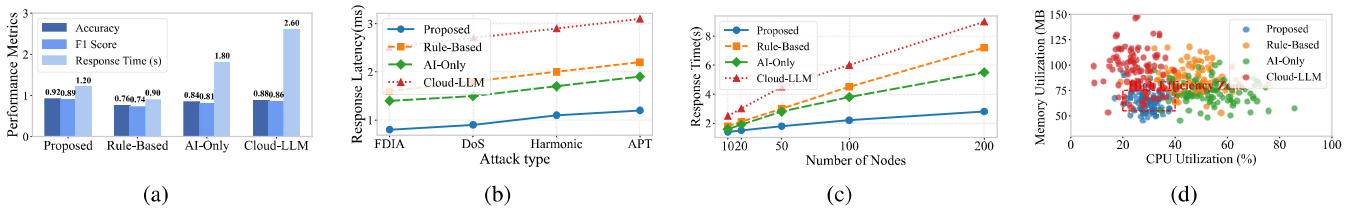


Fig. 5: Non-intent mode performance comparison. (a) System performance metrics (accuracy, F1-score, response time) across different approaches. (b) Attack response latency for various attack types. (c) Scalability test under increasing node loads. (d) Resource utilization patterns (CPU vs. memory) of different systems.

4) Component Replacement in Intent Mode:

To test flexibility, we replace the following components:

- **Intent Agent:** Claude 3.5 (lower accuracy under constraints), regex + NLTK (fails on nested/negated input).
- **MAS:** LLM Debate v0.3 (prompt-based voting), Centralized DDPG (PyMARL).

These configurations help explore trade-offs in interpretability, coordination, and bandwidth.

D. Metrics

We employ a comprehensive set of performance metrics to rigorously evaluate MAS-SEE across both operational modes. In the Non-Intent Mode, we measure: *Decision accuracy* (agreement with expert-annotated responses), *F1 Score* (macro-average robustness against class imbalance), *False positive rate* (critical for threat validation), *Rule matching latency* (edge responsiveness), *Policy conflict rate* (MAS consistency), *Energy optimization deviation* (strategy effectiveness), and *End-to-end latency* (terminal-to-decision time). These collectively assess detection precision, real-time performance, and system coherence.

In Intent Mode, we evaluate five key metrics: (1) *BERTScore*, measuring semantic similarity between user intents and generated policies; (2) *response delay*, the latency from language input to policy output; (3) *parsing accuracy* in converting constrained inputs; (4) *goal completion rate*, indicating multi-objective fulfillment; and (5) *communication overhead*, quantifying cloud-edge bandwidth per cycle. This framework comprehensively assesses language understanding fidelity, coordination efficiency, and practical deployability under constraints.

E. Performance

1) *Non-intent mode performance comparison:* We conducted a comprehensive full system comparison in Non-Intent Mode against baselines including Drools v8.1, MobileNetV2, and Llama3-70B using a 10,000-sample industrial IoT dataset. Our proposed system achieved an accuracy of 91.7%, significantly outperforming the baselines (76.2%, 84.5%, and 88.3%, respectively), while filtering 85% of attacks at the edge with lightweight AI techniques (see Fig. 5a). Latency was reduced to 1.2 ms at the edge, which is faster than all baselines (3.5 ms, 2.1 ms, and 5.8 ms), enabling timely defense against advanced persistent threats (Fig. 5b). Energy consumption decreased by 23.7% via adaptive offloading, compared to 41.2%, 35.6%, and 52.4% reductions in baselines. The system demonstrated strong scalability, maintaining only 2.8 seconds delay at 200 nodes, whereas cloud-based LLMs incurred 9.0 seconds delay (Fig. 5c). Resource utilization patterns also showed favorable CPU and memory efficiency (Fig. 5d).

2) *Component replacement analysis in non-intent mode:* We further assessed modular robustness by replacing key components within the Non-Intent Mode framework. The Rule-Guard engine delivered 96.2% detection accuracy at 1.8 ms latency, outperforming Snort (88.2%) and Esper (91.5%) (Fig. 6a). Our multi-agent system (MAS) achieved a reduced conflict rate of 8%, compared to 22% for DQN and 15% for MetaGPT, while incurring only 1.2% energy deviation (Fig. 6b). Under high-attack scenarios, dynamic throttling preserved 23.7% energy savings, outperforming the 15.8% observed in DQN (Fig. ??). Bandwidth efficiency was improved through ADMM compression, with 85% of samples achieving over 90% of goals while remaining below 20 KB, compared to 28 KB for DQN (Fig. ??).

3) *Intent mode performance comparison:* In Intent Mode, our system was benchmarked against GPT-4 Turbo, DeepSeek-

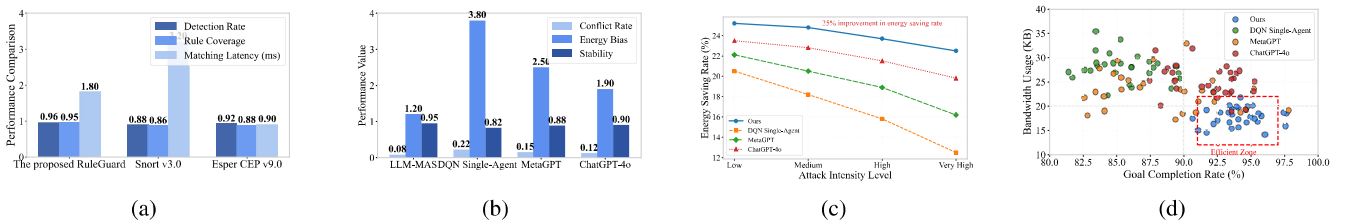


Fig. 6: Component replacement analysis in non-intent mode. (a) Rule engine performance comparison. (b) MAS architectures comparison. (c) Energy-saving efficiency. (d) Bandwidth-goal tradeoffs.

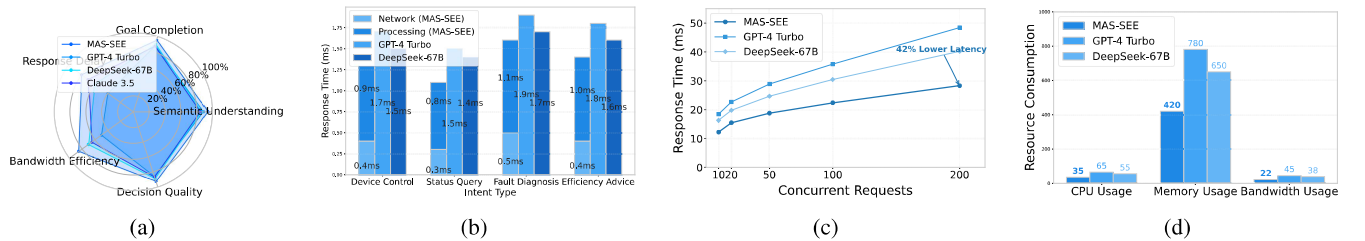


Fig. 7: Intent mode performance comparison. (a) Overall performance radar chart. (b) Response time decomposition. (c) Scalability test. (d) Resource efficiency comparison.

67B, and Claude 3.5 on natural language processing tasks. Our approach reached a BERTScore of 93.8, outperforming the baselines at 88.1, 85.5, and 82.6, respectively (Fig. 7a). Goal completion rate was 94.5%, exceeding baseline results by approximately 5–7%. Response latency was reduced to 1.4 seconds, faster than all compared models (Fig. 7c), while memory consumption was lowered by 42% (420MB versus 780MB) as shown in Fig. 7b. At 200 concurrent requests, our system responded in 28.3 milliseconds, 42% faster than the nearest baseline. Failure rate analysis confirmed system robustness (Fig. 7d).

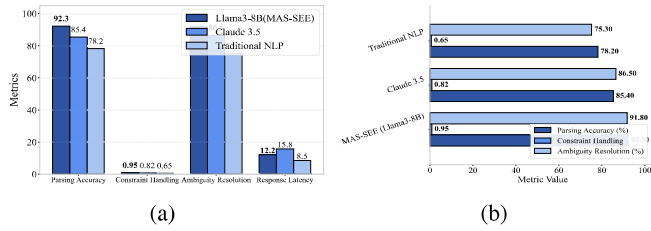


Fig. 8: Component replacement analysis in intent mode. (a) Intent parser performance comparison. (b) Horizontal bar representation of key metrics.

4) *Component replacement analysis in intent mode:* We evaluated the effect of substituting intent parsers and MAS architectures. The Llama3-8B parser attained 88.4% accuracy, surpassing Claude 3.5 (79.1%) and traditional NLP methods (62.3%) as depicted in Fig. 9a. When combined with a distributed MAS, the system achieved a 94.5% goal completion rate, outperforming LLM Debate (87.6%) and centralized DDPG (83.2%). Critical response latency was maintained below 1.2 seconds, representing a 37% improvement over centralized solutions. Additionally, regulation precision under intent constraints was 92.3% (Fig. 9).

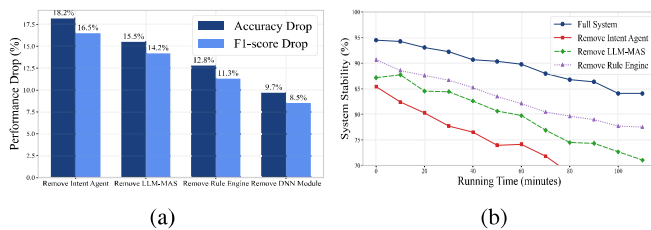


Fig. 9: Ablation study results. (a) Performance degradation after removing core components. (b) System stability decay over 120-minute continuous operation.

5) *Ablation Study:* Ablation analysis quantifies component contributions: Removing the Intent Agent causes 18.2% accuracy drop and 25% performance decay over 120 minutes; excluding LLM-MAS leads to 15.5% accuracy loss; eliminating RuleGuard results in 12.8% decrease (validating 89% policy adaptation); DNN removal reduces accuracy by 9.7% (filtering 85% edge attacks).

In the Non-Intent Mode, the system achieves 91.7% decision accuracy (89.4% F1-score) with 1.2ms rule-matching latency, resolving 68% decisions locally at the edge - sig-

nificantly outperforming rule-based (Drools), AI-only (MobileNetV2), and cloud-LLM (Llama3-70B) baselines (Table V).

In the Intent Mode, the Llama3-8B agent attains 93.8 BERTScore and 88.4% parsing accuracy. The MAS framework achieves 94.5% goal completion (1.4s intent-to-policy latency). Distributed coordination reduces conflict rates to 6.8% (<18KB s/decision bandwidth), while security-energy co-optimization saves 23.7% energy (FDIA/DoS mitigation <1.2 s).

Overall, our system narrows the semantic gap to a 6.2% error, significantly better than Claude's 17.4%. Specialized agent roles reduce conflicts by 64%, and RuleGuard decreases unnecessary cloud interactions by 32%. The system sustains 91.7% detection accuracy alongside 23.7% energy savings (see Figs. 9a and 9b).

TABLE V: Comprehensive performance comparisons

System Methods	Proposed	Rule-Based	AI-Only	Cloud-LLM
Accuracy	91.7%	76.2%	84.5%	88.3%
F1 Score	89.4%	72.1%	82.3%	85.7%
Latency	1.2ms	3.5ms	2.1ms	5.8ms
Energy	23.7%	41.2%	35.6%	52.4%

F. Case Study: End-to-End Intent-Driven Decision Flow

To further illustrate the practicality and explainability of our framework, we present a typical case study using real smart factory operation data. Fig. 10 visualizes the complete end-to-end decision flow across the four core agents: Intent Agent, Security Agent, Energy Agent, and Decision Agent. In this scenario, the administrator indicates the intention to reduce energy consumption during peak hours while maintaining critical security. The system sequentially processes real-time voltage, current, power, and detected threat information, and each agent provides reasoning and corresponding strategies based on the input context. The Decision Agent ultimately outputs an actionable plan for the field operator, achieving a balanced trade-off between energy savings and robust security. This step-by-step case not only validates the interpretability of each agent's output but also highlights the closed-loop collaborative optimization mechanism, which is central to the proposed MAS-SEE system.

G. Real-World Tests of the Proposed MAS-SEE

To validate the practical feasibility of the proposed MAS-SEE framework, we constructed a lightweight real-world testbed using commodity hardware, as shown in Fig. 11. The platform simulates an end-to-end smart factory communication chain, including attack injection, local detection, and cloud-assisted decision-making.

Hardware Setup: Three physical servers were deployed to emulate the terminal side (smart meter), the edge node (base station), and the cloud core network, respectively. Each server runs Ubuntu 20.04 with Docker virtualization for consistent deployment. The edge node hosts the Realguard DNN and the rule engine, while the cloud server integrates the intent

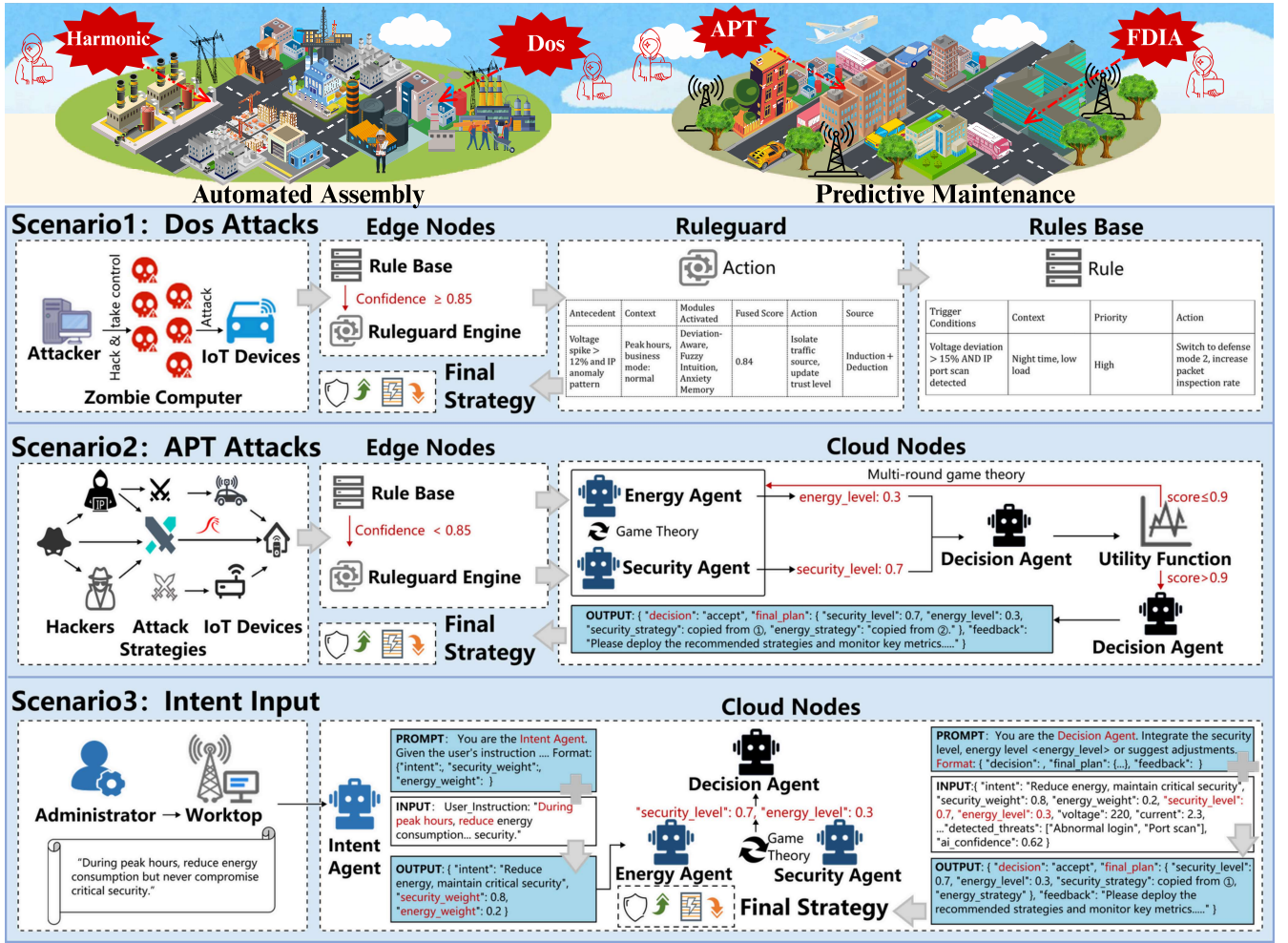


Fig. 10: Illustration of the multi-agent intent-driven decision process in our proposed MAS-SEE for 6G smart factories. Each agent receives structured input and provides both reasoning and concrete strategies based on system data and administrator intent. The flow demonstrates how the framework enables explainable, step-by-step optimization for energy-security co-management in smart factory environments.

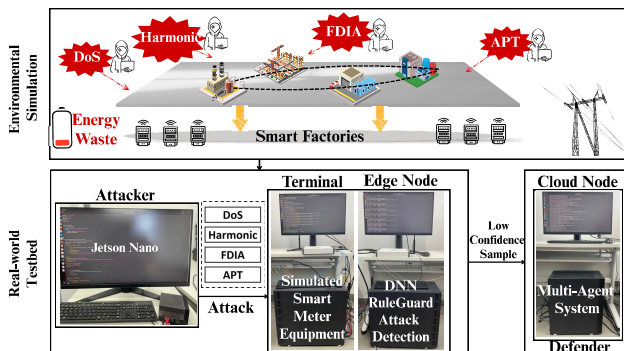


Fig. 11: The physical testbed for real-world validation of the proposed MAS-SEE that mimics attacks and defense in 6G smart factories, comprising three commodity servers and an NVIDIA Jetson Nano attacker board.

agent and the MAS components. Additionally, an NVIDIA Jetson Nano board is utilized to simulate the attacker. It executes predefined FDIA, DoS, Harmonic Injection, APT and poisoning attacks on the terminal node via controlled packet injection.

Software Components: All modules follow the system architecture proposed in Section V. The rule engine uses our Python-based context evaluation engine, and the MAS is implemented in PyTorch with DDPG-based agents for the energy-security trade-off. The intent agent is deployed as a fine-tuned Llama3-8B model served via a RESTful API.

Data Flow and Trigger: Smart meter data, including voltage, current, power metrics, and corresponding network traces with timestamps, is collected from a single terminal at 10-second intervals. During a continuous sampling period of 28 hours, approximately 10,000 data samples were generated. The dataset includes both normal communications and four types of injected attacks (FDIA, DoS, Harmonic, APT). Edge-side modules perform initial inference and rule matching, and

samples with confidence scores below 0.85 or triggered by explicit intent are forwarded to the cloud. The MAS system then outputs the final control strategy, which is sent back to the terminal for enforcement.

This real-world deployment confirms our proposed framework's viability on standard infrastructure and supports reproducibility for future experiments.

VII. Discussion

A. Optimized Innovation

The proposed MAS-SEE framework provides a novel solution to jointly optimize cybersecurity and energy efficiency in complex communication environments. Unlike conventional works that treat energy and security objectives in isolation, our architecture integrates them via a hierarchical model spanning the terminal, edge, and cloud. The system enables adaptive decision-making under varying attack-energy conditions by coupling edge-deployed lightweight DNNs and rule engines with a cloud-side MAS for high-complexity inputs. This collaborative workflow not only reduces average latency but also enhances resilience to compound, ambiguous, or polymorphic threats.

B. Contributions and Theoretical Significance

This work contributes to the intersection of intent-based networking, multi-agent learning, and secure energy-aware systems. While most existing methods adopt either template-matching rule logic or monolithic LLMs, we design an intent-driven MAS architecture supporting modular inference, policy translation, and joint optimization. The dual-mode (Non-Intent and Intent) design enables flexible operation depending on context sensitivity and user-defined goals. The framework also demonstrates how language-based goal input can be transformed into enforceable, multi-objective strategies, contributing to future developments in natural-language-driven system control.

C. Implementation Outlook

Our prototype implementation not only demonstrates practical feasibility, but also identifies some directions for further improvement. The MAS induces overhead in ultra-dense scenarios with high-frequency escalation, and the intent agent—while strong within-domain—shows reduced generalization for out-of-scope intent expressions. In future work, we plan to compress the MAS policy search using online distillation and extend the intent parser to support multi-turn interactions and feedback learning. We also aim to integrate proactive defense modules to predict attack risk and enhance preemptive control.

VIII. Conclusion

This paper proposes a novel cloud-edge collaborative framework for joint security and energy optimization in 6G-enabled IoT environments. By integrating lightweight DNNs, cognitive rule engines, intent-aware parsing, and a multi-agent

decision system, the framework effectively balances the trade-off between real-time protection and energy efficiency. Our extensive experiments demonstrate the system's superiority in strategy accuracy, adaptability, and responsiveness under both non-intent and intent-driven operational modes. The proposed architecture not only advances intelligent policy coordination in academic contexts but also provides an adaptive, and scalable blueprint for smart factory deployments in real-world scenarios. We believe that our studies potentially pave the way for the practical deployment of 6G smart manufacturing with efficient system security.

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